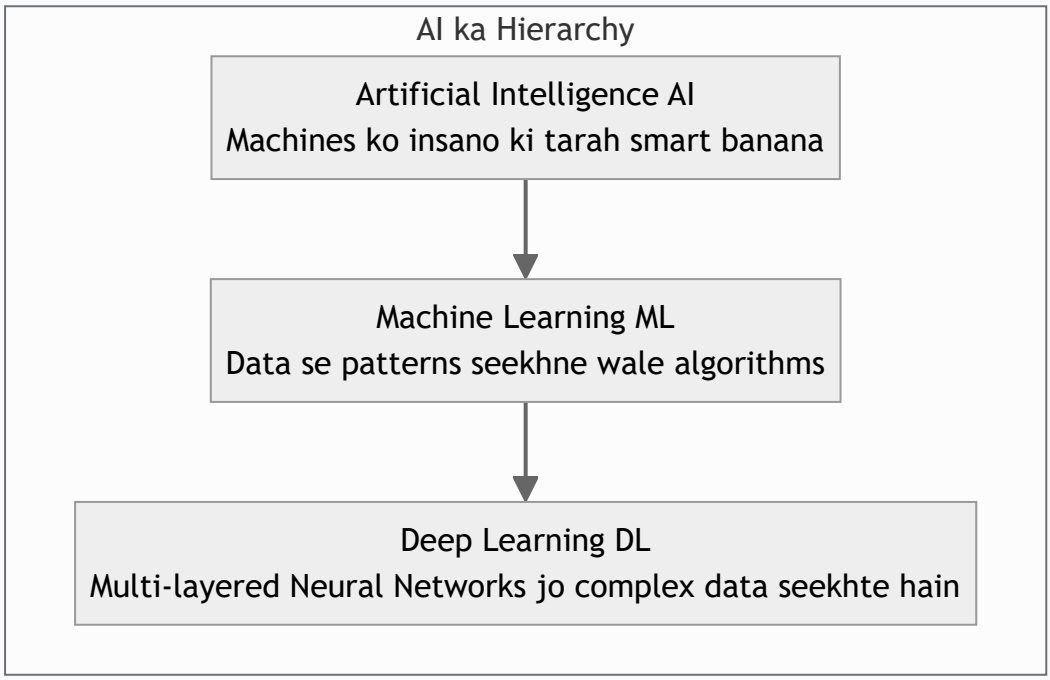
Submodule 36.01: Deep Learning kya hai? (Hinglish)

1. Deep Learning ki Paribhasha

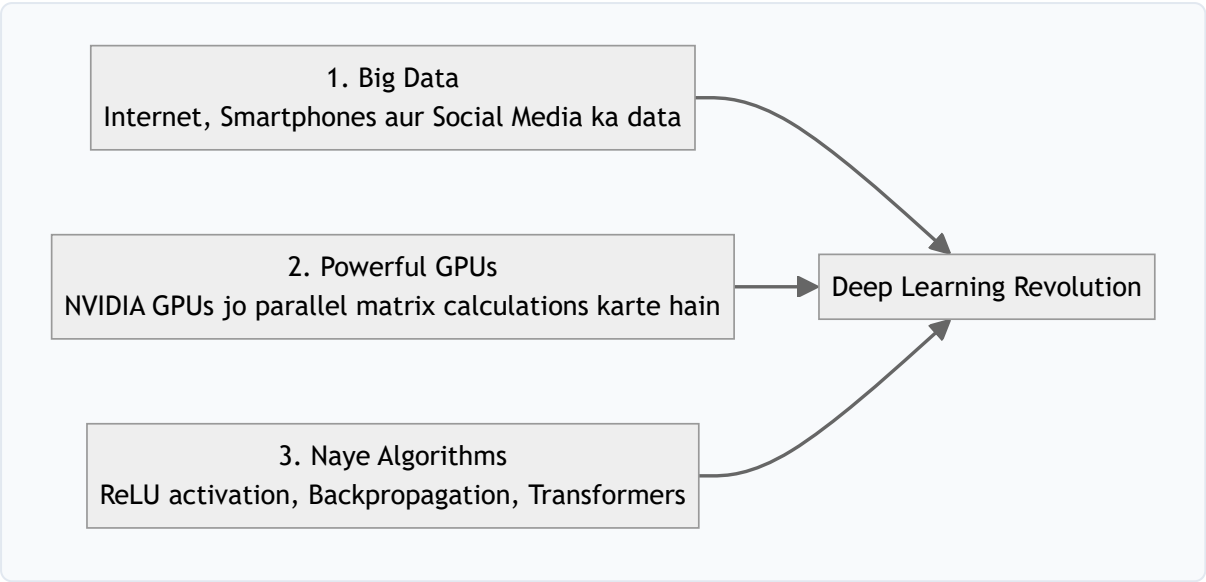
Deep Learning (DL) dar-asal **Machine Learning (ML)** ka ek advanced subset hai, jo khud **Artificial Intelligence (AI)** ke andar aata hai. Deep Learning human brain (insani dimaag) ke neurons aur neural networks ke structure se inspire hokar banaya gaya hai.



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2. Deep Learning Achanak Itna Popular Kyun Hua?

Neural networks ki theory 1950s mein hi invent ho chuki thi, lekin pichle 10-15 saalon mein Deep Learning ne poori duniya badal di. Iske piche teen bade factors hain:



1. **Big Data ki Uplabdhta**: Internet aur smartphones ki wajah se aaj hamare paas billions of images, videos, audios aur text documents maujood hain.
2. **GPU aur Compute Power**: Neural network mein lakho-crodon matrix multiplications hote hain. NVIDIA ke modern GPUs in calculations ko CPU ke mukable hazaron guna fast process kar sakte hain.
3. **Advanced Algorithms**: Naye activation functions (jaise **ReLU**) aur optimizers (jaise Adam) ne puraani training problems ko solve kar diya.

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3. Real-World Applications

Field	Real-Life Example	Deep Learning Technology
Computer Vision	Self-driving cars (Tesla), Medical X-ray aur MRI scan analysis	Convolutional Neural Networks (CNN)
Natural Language Processing	ChatGPT, Google Translate, AI Chatbots	Transformers aur Attention Models
Voice & Speech	Siri, Alexa, Speech-to-text transcription	Recurrent Neural Networks (RNN)
Recommendation Engines	YouTube videos aur Netflix movies ki recommendation	Deep Neural Collaborative Filtering
Generative AI	AI image generation (Midjourney), AI video generation	Diffusion Models aur GANs

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4. Visual Reference

